



FIRST SEMESTER 2026-27
COURSE HANDOUT

Date: 01.08.2026

In addition to part I (General Handout for all courses appended to the Time table) this portion gives further specific details regarding the course.

Course No : EEE G593
Course Title : Power Quality
Instructor-in-Charge : Ashish Patel
Instructor(s) : Lecture: Ashish Patel, Practical: Vedant Vyas.

1. Course Description: This ME course is meant for equipping students with knowledge of prevalent power quality issues in the modern power distribution system, their causes and classifications covering voltage and current based power quality problems such as sag, swell, unbalance, harmonics and poor power factor. Various standards on power quality and their monitoring techniques are also covered. The course also majorly deals with power quality compensation techniques including both passive and active compensation, their design, analysis and modeling. The control techniques for active power compensators and active filters are also discussed. The lab component is designed to enrich students' knowledge of power quality with practical hands on and simulation exercises.

2. Scope and Objective of the Course: A ME course focusing on electric power quality typically covers the analysis, mitigation, and improvement of power quality issues, including voltage variations, harmonics, and other disturbances, preparing graduates for careers in power utilities and industries.

The objectives of this course are designed to help students to:

- Understand various power quality issues and their impact.
- Learn about power and power factor in single-phase and three-phase systems.
- Study conventional and active compensation techniques for power factor correction and load voltage regulation
- Gain knowledge of active filter techniques for harmonics elimination.
- Understand power quality improvement in power supplies, drive systems, and renewable energy systems.

Learning outcomes of the course have been incorporated in the course plan on next page.

3. Text Books:

- T1:** B. Singh, A. Chandra, K.A. Haddad, "Power Quality: Problems and Mitigation Techniques", Wiley, 2015, 1st Edition.
- T2:** R. C. Dugan, M. F. McGranaghan, S. Santoso, H. W. Beaty, "Electrical Power Systems Quality", McGrawHill, 2012, 3rd Edition.

4. Reference Books:

- R1:** Ned Mohan, T.M. Undeland, W.P. Robbins, "Power Electronics", Wiley, 2022, 3rd Edition.
- R2:** E. F. Fuchs, M.A.S. Masoum, "Power Quality in Power Systems and Electrical Machines", Elsevier Inc., 2008, 1st Edition.
- R3:** C. Sankaran, "Power Quality", CRC Press, 2002, 1st Edition.



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5. Course Plan:

Module No. & Name	Lecture Sessions	References	Learning Outcomes
1. Fundamentals of Power Quality	L1-L5	T1 (1.1-1.6, 2.1-2.6), T2 (1.1-1.4, 2.1-2.12)	Introduction to power quality problems related to voltage variations, waveform distortion, low power factor etc., their causes and effects. Power quality terminologies and definitions, industry standards, monitoring.
2. Non-linear Loads and Harmonics	L6-L9	T1 (7.1-7.6), T2 (5.1-5.10), R1 (3.2)	Introduction to non-linear loads and problem of harmonics caused by them, their analysis, modeling and simulation. Calculation of active & reactive powers and power factor for non-linear loads
3. Passive Compensation Techniques	L10-L12	T1 (3.1-3.6)	Introduction to passive shunt and series compensators, their classification, principle of operation, analysis, design and simulation.
4. Active Shunt Compensation Techniques	L13-L18	T1 (4.1-4.6)	Introduction to shunt active compensators- DSTATCOMs, their classification, principle of operation, analysis, design, modeling control techniques and simulation.
5. Active Series Compensation Techniques	L19-L22	T1 (5.1-5.6)	Introduction to series active compensators- DVRs, their classification, principle of operation, analysis, design, modeling control techniques and simulation.
6. Unified Power Quality Compensators	L23-L26	T1 (6.1-6.6)	Introduction to unified compensators- combining shunt and series active compensators (UPQC), their classification, principle of operation, analysis, design, control techniques and simulation. Advantages and disadvantages of UPQC.
7. Passive power filters	L27-L29	T1 (8.1-8.8), T2 (6.1-6.8)	Passive power filters for harmonic mitigation, their classification, principle of operation, analysis, design, modeling and simulation. Problem of parallel resonance in passive filters and its mitigation
8. Active Power Filters	L30-L36	T1 (9.1-9.6, 10.1-10.6)	Introduction to shunt and series active power filters for harmonic mitigation, their classification, principle of operation, analysis, design, modeling, control techniques, and simulation. Introduction to hybrid active power filters.
9. Power Quality Applications and Related Topics	L37-L40	T2 (9.3-9.8, 10.1-10.5), class notes.	Power quality issues and solutions in areas like DC power supplies, renewable energy systems, industrial applications, electric vehicle charging stations and smart grids.



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6. Evaluation Scheme:

Component	Duration	Weightage (%) (total marks-100)	Date & Time	Nature of component (Close Book/ Open Book)
Mid-Semester Test	90 Min.	25%	10/10 AN2	CB
Comprehensive Examination	3 hours	35%	16/12 AN	CB (15 marks) + OB (20 marks)
Seminar	10 min.	10%	Will be announced	OB
Lab	90 Min.	20%	Regular lab sessions	OB
Assignment (Take home, Simulation)	-	10%	Will be announced	OB

7. Grading & Minimum passing marks: Grading is relative as per institute policy. Mid-semester grades will include all components till mid-semester exam. A student should obtain 25% of the average marks of the Class, or 10% of the Course Total Marks, whichever is lower to clear the course.

8. Chamber Consultation Hour: Saturday, 11 AM- 12 PM, 2210/O.

9. Notices: Notices will be posted on Nalanda.

10. Make-up Policy: It will be given on extremely genuine grounds.

11. List of Experiments:

- Measurement of real, reactive and apparent powers and power factor in AC circuits (Hardware using power analyzer or wattmeter & voltage-current meters)
- Analysis of non-linear waveforms & associated harmonics, and measurement of THD using power analyzer or power-analysis enabled oscilloscope.
- Modeling and simulation of non-linear and reactive power loads and the calculation of power factor and THD in simulation environments like MATLAB/PSIM.
- Design, analysis and simulation of passive shunt compensator.
- Design, analysis and simulation of passive series compensator.
- Design, analysis, control and simulation of active shunt compensator (DSTATCOM).
- Design, analysis, control and simulation of active series compensator (DVR).
- Design, analysis, and simulation of shunt passive power filter for harmonic elimination.
- Design, analysis, control and simulation of shunt active power filter (shunt APF) for harmonic elimination.
- Estimation of power losses in an active power filter using simulation based study.

Instructor-in-charge
Course No. EEE F312